

Y29&Y34 of LCD Protocol

This protocol is a programmatic application protocol for the transmission of operating status and faults between the LCD and the electric vehicle controller.

1. Adopting the international standard SIF communication protocol, the interface is universal and convenient.

2. The master-slave mode adopts single-wire unidirectional transmission, that is, only one transmission thread is needed, the electric vehicle controller is the sender, and the multi-function display The indicator is the receiver.

3. The transmission baud rate has a wide adaptive range, and the host can use the idle time to send data.

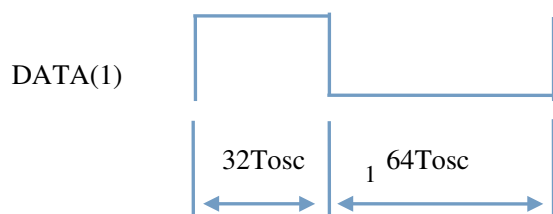
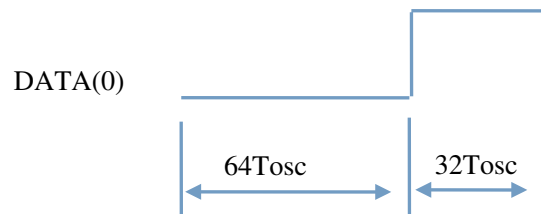
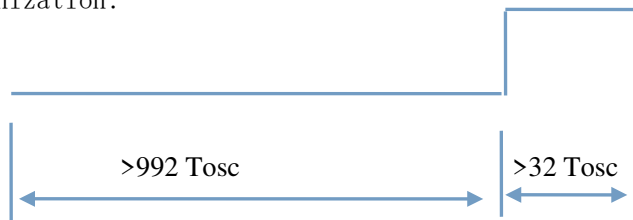
4. One frame of data is transmitted at a time, including 97 Bits: a start bit, 12x8 data bits, the line idle status bit is required to be low after the transmission is completed.

5. $32\mu s < T_{osc} < 320\mu s$.

6. The level of the data complies with the TTL specification.

	Synchronize	DATA0	DATA1	DATA2-DATA10	DATA11
Signal	Low High Level	8 bits Low high level	8 bits Low high level	8 bits Low high level	8 bits Low high level
Content	Meaningless	Equipment code 8 bits	serial number bits 8	Data bits 8	Checksum 8bits
Name		Device_Code	Seq_Code		checksum

Synchronization:



7. Frame format

Name	Start sync signal	Device_Code	Seq_Code	Data content	Checksum
Content		DATA0	DATA1	DATA2-DATA10	DATA11
Length		1Byte	1Byte	9Byte	1Byte
Note		Fixed Fill 0x08	Fixed Fill 0x61	Detailed description of data content	Detailed description of data content

8. Detailed description of the data content (if the specified state exists, the specified Bit position is 1, otherwise it is set to 0, and the reserved items are filled with the value 0);

DATA2					
D7	Standby				
D6	Standby				
D5	Standby				
D4	Standby				
D3	Side support instructions				
D2	Standby				
D1	Parking instructions				
D0	Standby				
DATA3					
D7	6KM push cart logo				
D6	Hall fault				
D5	Handlebar failure				
D4	Controller fault				
D3	Undervoltage protection				
D2	Cruise				
D1	Boost				
D0	Motor phase loss				
DATA4					
D7	Fourth gear indicator				
D6	Motor running (1 running 0 stop) pwm output or not				
D5	Brake				
D4	Controller protection				
D3	Coasting charging				
D2	Anti-skid protection				
D1	Three speed	1	1	0	0
D0		1	0	1	0
		High speed	Medium speed	Low speed	No 3-speed controller

DATA5	
D7	70% current mark
D6	Enable push-to-talk
D5	Enable EKK standby power
D4	Overcurrent protection
D3	Locked-rotor protection
D2	Reverse
D1	Electronic brakes
D0	Speed limit
DATA6	
D7-D0	Running current (unit A) (The high bit of negative current is 1 The high bit of positive current is 0)
DATA7	
D7-D0	Speed double byte, high byte, the number of three Hall changes within 500 milliseconds
DATA8	
D7-D0	Speed double byte, low byte, the number of three Hall changes within 500 milliseconds
DATA9	
D7-D0	Battery capacity: 0-100% current ratio value
DATA10	
D7-D0	Current: 0-100% current ratio

Protocol description:

1. Device code Device_Code is a constant constant, LCD device code is DATA0 = Device_Code = 0X08;

2. The serial number Seq_Code is a constant constant, and the serial number code is DATA1 = Seq_Code = 0x61;

3. Checksum DATA11(8Bit), 8Bit exclusive OR value of DATA0-DATA10.
DATA11=

DATA0^DATA1^DATA2^DATA3^DATA4^DATA5^DATA6^DATA7^DATA8^DATA9^DATA10

4. Send in order according to the sending format
Sync code DATA0 ~ DATA11

5. The standard 0,1 high level time and low level time ratio are 1:2 and 2:1.

6. Speed information: The speed information includes the change information of the HALL sensor inside the motor. This parameter can get the current speed of the motor.

Degree value.

N: The number of halls that the motor has rotated per hour

P: the number of pole pairs of the current motor

D: The diameter of the current motor

V: current speed

$$V = (N * D * \pi) / 6P$$

The unit is converted into KM/H. In this protocol, the controller provides a hall number N_Tx that the motor has rotated within 500ms, then (N = N_Tx*7200); P is the number of motor pole pairs, D is the motor diameter, which needs to be provided by the vehicle manufacturer Or fill in an approximate value, such as 23 pairs of poles, 16 inches.

Vehicle status: fault information simulation conditions

Undervoltage: battery undervoltage

Controller failure: Power on after the power line and the motor line are short-circuited.

Motor Hall failure: unplug the Hall wire

Handle failure: unplug the handle

Brake failure: stop the car and power on

Cruise: At this time, the vehicle is in cruise mode

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